MILITARY SPECIFICATION

CRADLE, AIRCRAFT-MISSILE COMPONENTS MMU-1/E

This specification is approved for use by the Department of the Air Force, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

* 1.1 Scope. This specification covers a cradle for aircraft-missile components, designated MMU-1/E.

2. APPLICABLE DOCUMENTS

* 2.1 Government documents.

* 2.1.1 Specifications and standards. Unless otherwise specified (see 6.2), the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specification and Standards (DoDISS) specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-P-116	-	Preservation, Method of.
DOD-D-1000	-	Drawing, Engineering and Associated List.

STANDARDS

FEDERAL

FED-STD-595	-	Colors

MILITARY

MIL-STD-105	-	Sampling Procedures and Tables For Inspection By Attributes.
MIL-STD-129	-	Marking For Shipment and Storage.
MIL-STD-130	-	Identification Marking of U.S. Military Property

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: San Antonio ALC/MMEDO, Kelly AFB, Texas 78241 by using the self addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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* MIL-STD-794 - Part and Equipment, Procedures for Packaging and Packing.

* MIL-STD-1595 - Qualification of Aircraft, Missile and Aerospace Fusion Welders.

DRAWINGS

Air Logistics Corporation

101719 - Adapter, Component Installation Trailer, Model 3500

Air Force

* 7545352 - Requirements for Finishes, Protective and Codes for San Antonio ALC Ground and Ground Support Equipment.

* (Copies of specifications, standards and drawings required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.1.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 First article. This specification makes provisions for first article testing.

3.2 Design and construction.

3.2.1 The cradle and component parts thereof, individually and grouped, shall inherently provide for maximum reliability commensurate with its intended use (see 6.1) and other criteria specified herein.

3.2.2 Assembly drawing. The requirements of Drawing 101719 apply as requirements of this specification with the exceptions and additions specified herein. When the drawing and this specification conflict, this specification shall govern (see 6.2).

3.2.3 Commercial items. Components used in lieu of commercial parts specified on the drawing shall be functionally and dimensionally interchangeable with the parts specified and must be approved by the procuring activity.

3.2.4 Welding. Welding shall be accomplished by welders certified in accordance with MIL-STD-1595.

3.3 Performance. The cradle shall withstand a proofload of 200 percent of the working load of 8,000 lbs. The cradle shall be designed for a fore, aft, and side load of 4,000 lbs.

* 3.4 Interchangeability. All parts having the same manufacturer's part number shall be functionally and dimensionally interchangeable. The drawing number requirements of DOD-D-1000 shall govern changes in the manufacturer's part numbers.

3.5 Finish. The cradle shall be finished in accordance with Air Force Drawing 7545352. The color shall be Color 24052 of FED-STD-595.

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3.6 Identification of product. Equipment, assemblies, and parts shall be marked for identification in accordance with MIL-STD-130.

3.7 Workmanship. The cradle, including all parts and accessories, shall be constructed and finished in a workmanlike manner. Particular attention shall be given to neatness and accuracy of marking of parts and assemblies; thoroughness of welding, brazing, painting, riveting, and machine screw assemblies, and freedom of parts from burrs and sharp edges.

3.7.1 Riveting. Riveting operations shall be carefully performed to insure that the rivets are tight and satisfactorily headed.

3.7.2 Multiple pass welds. When a weld is built up by two or more beads or passes, preceding beads shall be cleaned free of scale and flux by chipping or scraping and brushing.

4. QUALITY ASSURANCE PROVISIONS

* 4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of tests. The inspection and testing of the cradle shall be classified as follows:

a. Acceptance tests (see 4.3).

b. First article testing (see 4.5).

4.3 Acceptance tests. The acceptance tests shall consist of the following:

a. Individual test (see 4.3.1).

b. Sampling plan and test (see 4.3.2).

4.3.1 Individual test. Each cradle shall be subjected to the test specified in 4.4.1.

4.3.2 Sampling plan and test.

4.3.2.1 Lot. The lot definition, formation, and size shall be in accordance with MIL-STD-105.

4.3.2.2 The visual and dimensional inspection of components, assemblies, and end items, and the end items subjected to the test specified in 4.4.2, shall be selected for inspection in accordance with MIL-STD-105.

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4.4 Test methods.

4.4.1 Examination of product. The cradle shall be examined to determine compliance with the requirements specified herein with respect to material, dimensions, workmanship, marking, et cetera.

4.4.2 Proofload. The cradle shall be subjected to a proofload of 16,000 lbs applied symmetrically within the cradle. The proofload shall be applied by means of a cylindrical fixture or jig that will assume diameters of 15, 45, and 61 in. The fixture or jig shall so apply the load to the mounting pads that the pads assume the shape of the fixture or jig. The load shall be applied with the support pads in each of the three fixed positions. The cradle, supported by its four support faces, shall be tilted 10° longitudinally in each of the positions and the proofload applied. A 4,000-lb fore, aft, and side load shall be applied independently to the fixture or jig with an 8,000-lb load on the supports in all three fixed positions. Failure or permanent deflection of the cradle during the test shall be cause for rejection.

4.5 First article testing.

4.5.1 First article test sample. The first article test sample shall consist of one cradle representative of the production equipment. It shall be tested at a laboratory designated by the procuring activity or, when so specified in the contract (see 6.2), at the contractor's plant under the supervision of the procuring activity.

4.5.2 First article tests. First article tests shall consist of all the tests described under 4.4.

4.6 Inspection of the preservation, packaging, and packing. The inspection of the preservation, packaging, and packing shall be in accordance with the instructions in section 5 or the specifications referenced therein.

* 5. PACKAGING

* 5.1 Preservation, packaging, and packing. The cradles shall be prepared for delivery in accordance with MIL-STD-794. Preservation and packaging for level A shall be in accordance with method III of MIL-P-116.

5.2 Marking for shipment and storage. Interior packages and exterior shipping containers shall be marked in accordance with MIL-STD-129. The nomenclature shall be as follows:

CRADLE, AIRCRAFT-MISSILE COMPONENTS MMU-1/E

6. NOTES

6.1 Intended use. The MMU-1/E cradle covered by this specification is intended to be used for installing and removing cylindrical-shaped components from weapons systems and for handling wing and stabilizer sections.

6.1.1 This cradle is used with the ETU-8/E or ETU-9/E trailer and may be roller transferred to other trailers or stands. Four ETU-2/E rollers are used but not furnished with the cradle.

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6.2 Ordering data.

* 6.2.1 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. A statement to the effect that should it become necessary to change the drawings to achieve the performance requirements of the specification, the contractor will immediately notify the contracting officer with recommendations for drawing changes to obtain the required performance. To the extent approved, such changes shall be processed in accordance with the provisions of the changes clause of this contract.

c. Point of inspection (see 4.5.1).

d. Selection of applicable levels of preservation, packaging, and packing (see section 5.).

6.3 Changes from previous issue. The margins of this specification are marked with asterisks to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
	☐ USER
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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